

Predicting Realized Volatility with Machine Learning

A Data Science Project — Optiver Realized Volatility Dataset
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Overview

This project builds and evaluates a machine learning pipeline to predict short-term realized volatility using real order book and trade data from Optiver, a global market maker, covering 112 stocks across thousands of 10-minute trading windows. The core challenge was twofold: engineering informative features from irregular, high-frequency market microstructure data, and training a model whose predictions genuinely reflect what drives volatility rather than a coincidental correlation in the data.

Methodology

Feature engineering was carried out in SQL using DuckDB, aggregating raw order book and trade records into per-window measures of spread, price volatility, and order flow imbalance, with irregular timestamps handled through forward-filling rather than treated as missing data. A naive persistence baseline was established first, followed by a linear regression benchmark and a LightGBM model trained with a custom sample-weighting scheme to align its objective with the RMSPE metric used for evaluation throughout. SHAP was applied to confirm the final model's reasoning matched real market microstructure intuition rather than an arbitrary pattern in the data.

Key Results

Model	RMSPE
Naive Persistence Baseline	0.3414
Linear Regression	0.3933
LightGBM (Final Model)	0.2321

The final model improved on the naive baseline by roughly a third and landed within the 0.20–0.23 RMSPE range achieved by competitive solutions to the original Optiver competition. The linear regression notably underperformed the baseline, a result traced to an unfair training setup rather than a weakness in linear models themselves.

Notable Findings

Three features, current-window volatility, order book update frequency, and bid-ask spread, accounted for roughly 73% of the model's predictive weight, with SHAP confirming each drove predictions in the direction market microstructure theory would expect. Trade data proved far more complete than anticipated, with only 0.0044% of windows containing zero trades, meaning trade-derived features could be used across almost the entire dataset without heavy reliance on imputation.

Technical Skills Demonstrated

Python (pandas, NumPy, scikit-learn), SQL / DuckDB, LightGBM, SHAP, Git / GitHub.